1. Number of MAC per Layer

Layer 1 $784 \times 256 = 200,704$ MAC's

Layer 2 $256 \times 128 = 32,768$ MAC's

Layer 3 $128 \times 10 = 1,280$ MAC's

2 Total MAC's

$200,704 + 32,768 + 1,280 = 234752$ Total MAC's

3 Total no. of trainable Parameters

234752 weights

4 Total weight memory in bytes (FP32 4 bytes each)

$234,752 \times 4 = 939008$ bytes

5 Total activation memory in bytes

$(784 + 256 + 128 + 10) \times 4 = 4712$ bytes

6. Arithmetic Intensity

2 × total MAC's / (weight bytes + activation bytes)

$2 \times 234752 / (939008 + 4712)$

$469504 / 943720$

$= 0.4975$